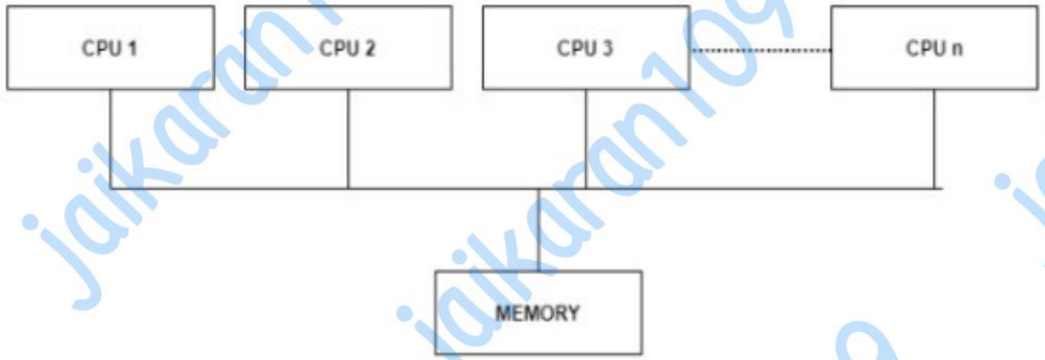
OUTLINE

- **M**ultiprogramming and Multiprocessing
- **P**arallel Processing
- **F**lynn's classification

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Multiprogramming and Multiprocessing:

- **Multiprocessor** systems have multiple processors working in parallel that share the computer clock, memory, bus, peripheral devices etc.



- To improve the performance of a CPU we have two options:

1) Improve the hardware by introducing faster circuits.
2) Arrange the hardware such that more than one operation can be performed at the same time.

Since, there is a limit on the speed of hardware and the cost of faster circuits is quite high, we have to adopt the 2nd option.

- ✓ **Multiprogramming** is a primary form of parallel processing in which several programs run at the same time on a uniprocessor system. The purpose of multiprogramming is to maximize CPU time.

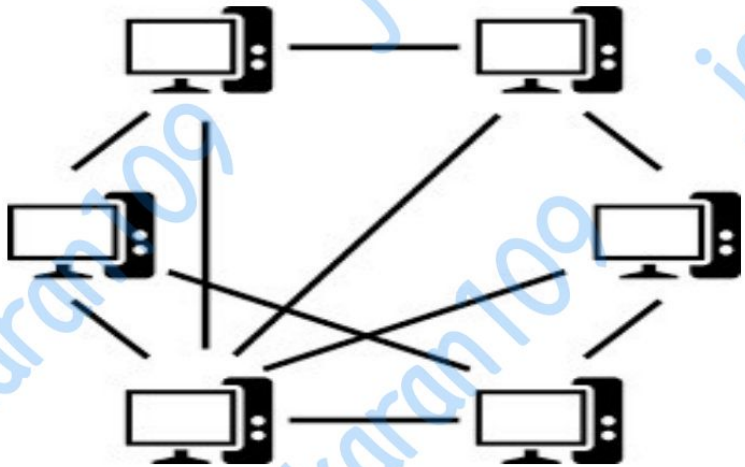
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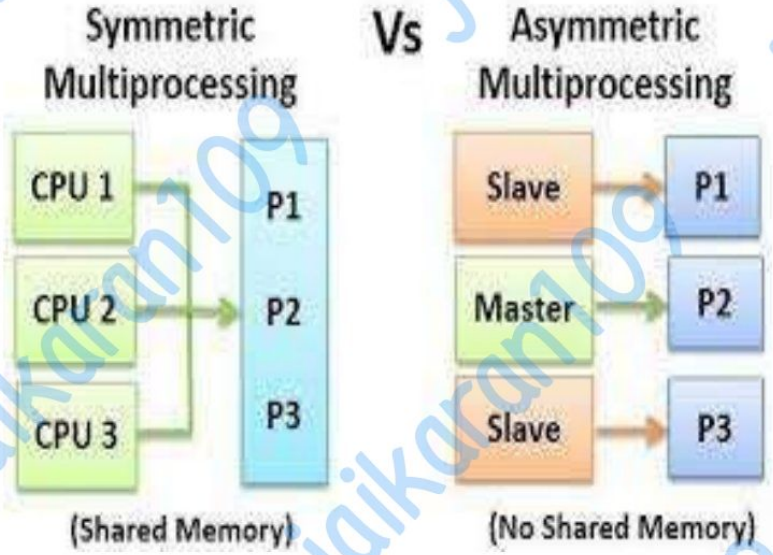
Multiprocessor Architecture:

Types of Multiprocessors:

- There are mainly two types of multiprocessors i.e. symmetric and asymmetric

Symmetric Multiprocessors: Each processor contains a similar copy of the operating system and they all communicate with each other. All the processors are in a peer to peer relationship i.e. no master-slave relationship exists between them.





Asymmetric Multiprocessors: In asymmetric systems, each processor is given a predefined task. There is a master processor that gives instruction to all the other processors. It contains a master slave relationship.

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Multiprocessor Architecture:

Advantages of Multiprocessor Systems:

- More reliable Systems: In a multiprocessor system, even if one processor fails, the system will not halt.
- Enhanced Throughput: If multiple processors are working in tandem (one behind another), then the throughput of the system increases i.e. number of processes getting executed per unit of time increase. If there are N processors then the throughput increases by an amount just under N.

Disadvantages of Multiprocessor Systems:

- Increased Expense: It is quite expensive. It is much cheaper to buy a simple single processor system than a multiprocessor system.
- Complicated Operating System Required: There are multiple processors in a multiprocessor system that share peripherals, memory etc. So, it is much more complicated to schedule processes and impart resources to processes than in single processor systems.
- Large Main Memory Required: All the processors in the multiprocessor system share the memory. So a much larger pool of memory is required as compared to single processor systems.

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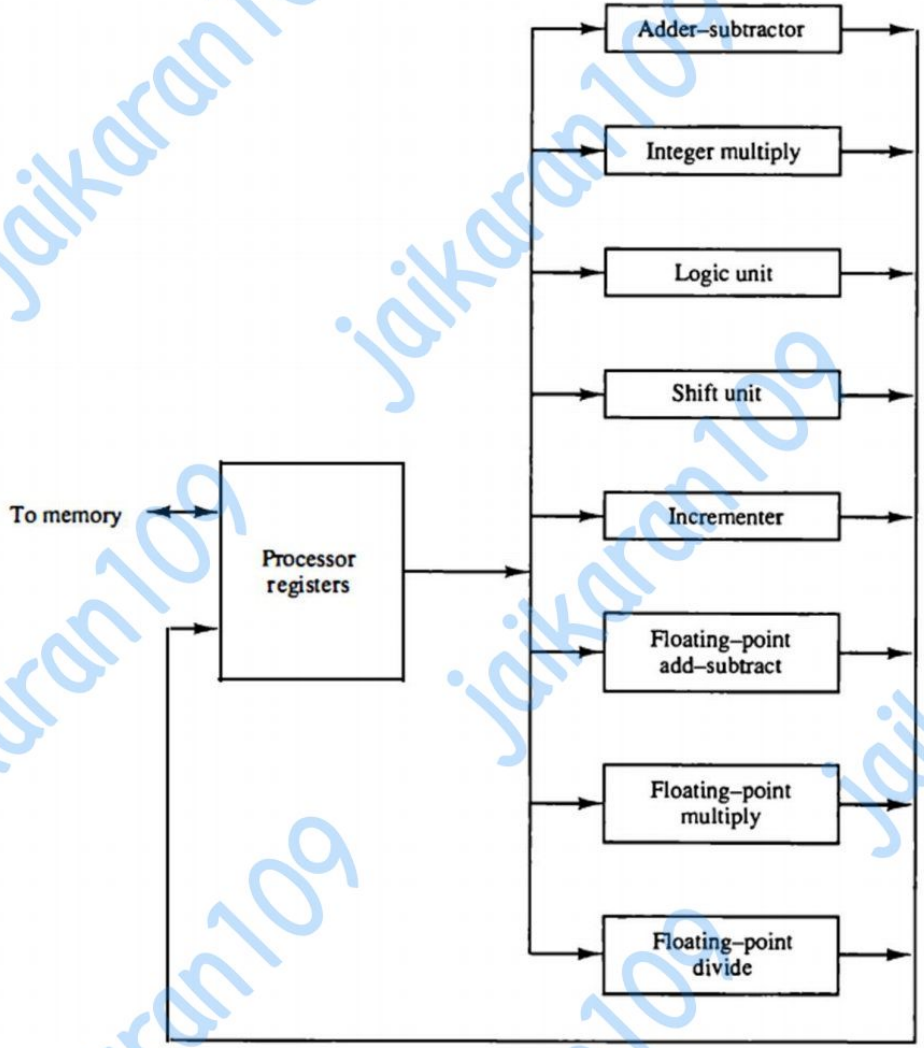
Parallel Processing:

- Parallel processing is used to provide simultaneous data-processing tasks for the purpose of **increasing the computational speed** of a computer system.
- Instead of processing each instruction sequentially as in a conventional computer, a parallel processing system is able to perform concurrent data processing to achieve **faster execution time**.
- For example, while an instruction is being **executed in the ALU**, the next instruction **can be read from memory**.
- The purpose of parallel processing is to **speed up the computer processing capability and increase its throughput**, that is, the amount of processing that can be accomplished during a given interval of time.
- The **amount of hardware increases** with parallel processing. and with it, the cost of the system increases. However, technological developments have reduced hardware costs to the point where parallel processing techniques are economically feasible.

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Parallel Processing:

- Below figure shows one possible way of separating the execution unit into eight functional units operating in parallel.



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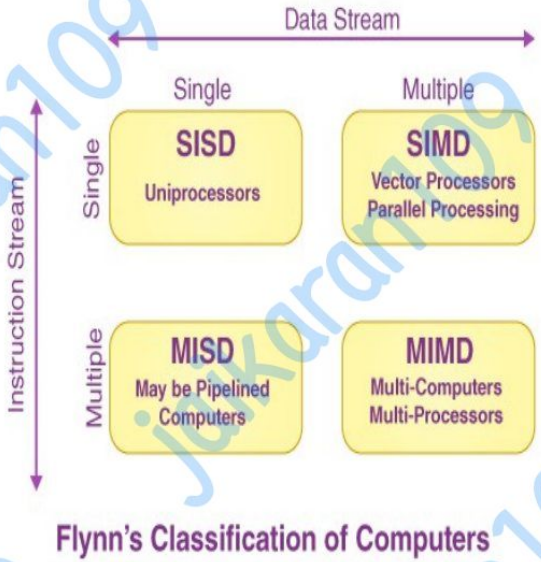
Parallel Processing:

- There are a variety of ways that parallel processing can be classified. It can be considered from the internal organization of the processors, from the interconnection structure between processors, or from the flow of information through the system.
- One classification introduced by **M. J. Flynn** considers the organization of a computer system by the **number of instructions and data items** that are manipulated simultaneously.
- The normal operation of a computer is to fetch instructions from memory and execute them in the processor.
- The sequence of instructions read from memory constitutes an **instruction stream.**
- The operations performed on the data in the processor constitutes a **data stream.**
- Parallel processing may occur in the instruction stream, in the data stream, or in both.

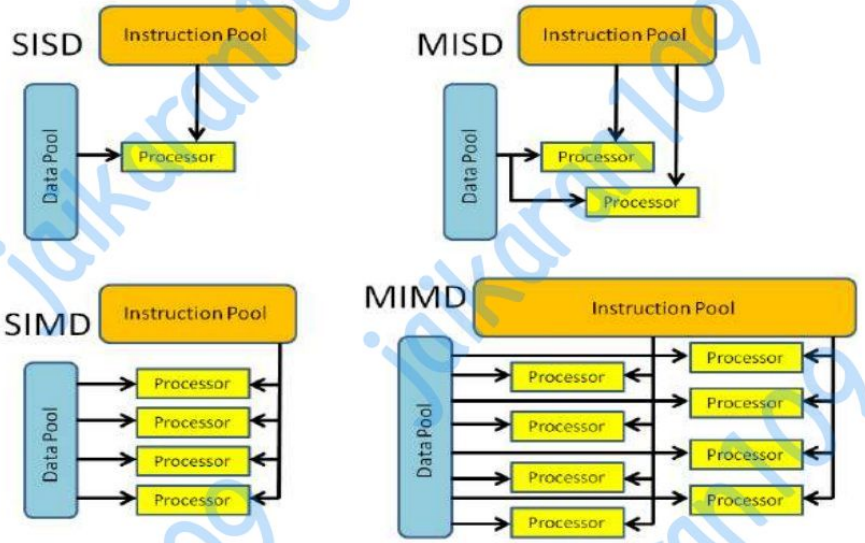
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Flynn's classification:

- Flynn's classification divides computers into four major groups as follows:
- **Single instruction stream, single data stream (SISD)**
- **Single instruction stream, multiple data stream (SIMD)**
- **Multiple instruction stream, single data stream (MISD)**
- **Multiple instruction stream, multiple data stream (MIMD)**



Flynn's Classification of Computers



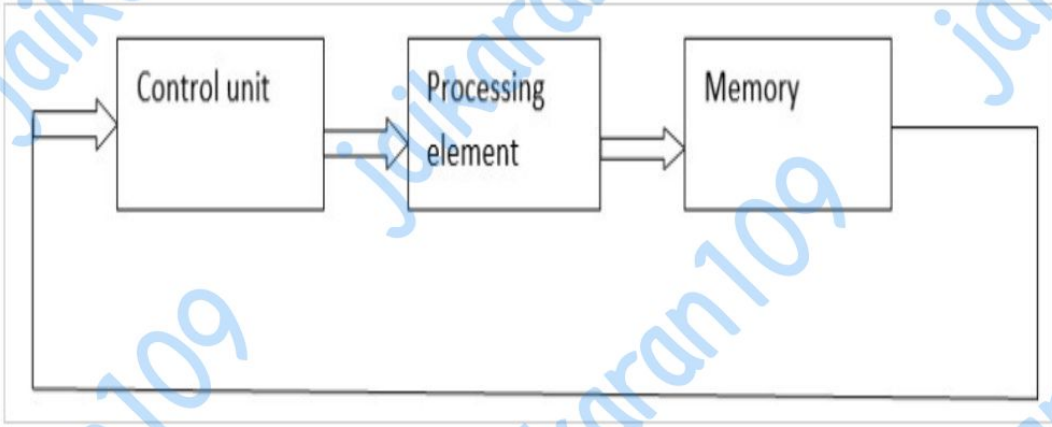
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Flynn's classification:

- **Single instruction stream, single data stream (SISD):**

Single instruction: Only one instruction stream is being acted or executed by CPU during one clock cycle.

Single data stream: Only one data stream is used as input during one clock cycle.



- ✓ A SISD computing system is a uniprocessor machine that is capable of executing a single instruction operating on a single data stream.
- ✓ Most conventional computers have SISD architecture where all the instruction and data to be processed have to be stored in primary memory.
- ✓ **Example**: Von Neumann types of computers.

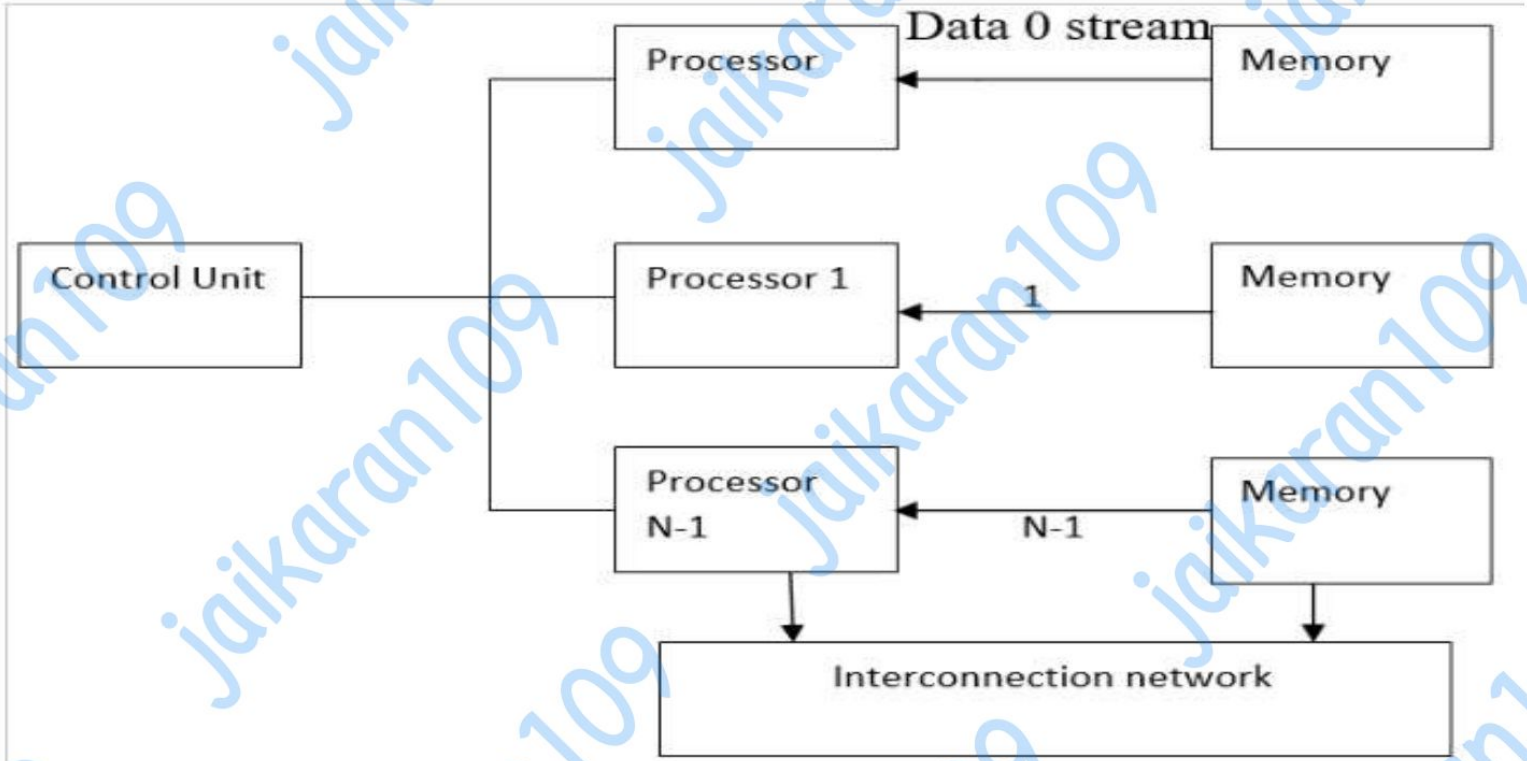
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Flynn's classification:

- **Single instruction stream, multiple data stream (SIMD):**

A SIMD system is a multiprocessor machine, capable of executing the same instruction on all the CPUs but operating on the different data stream. **Example**: Vector computers, array computers

IBM 710 is the real life application of SIMD.

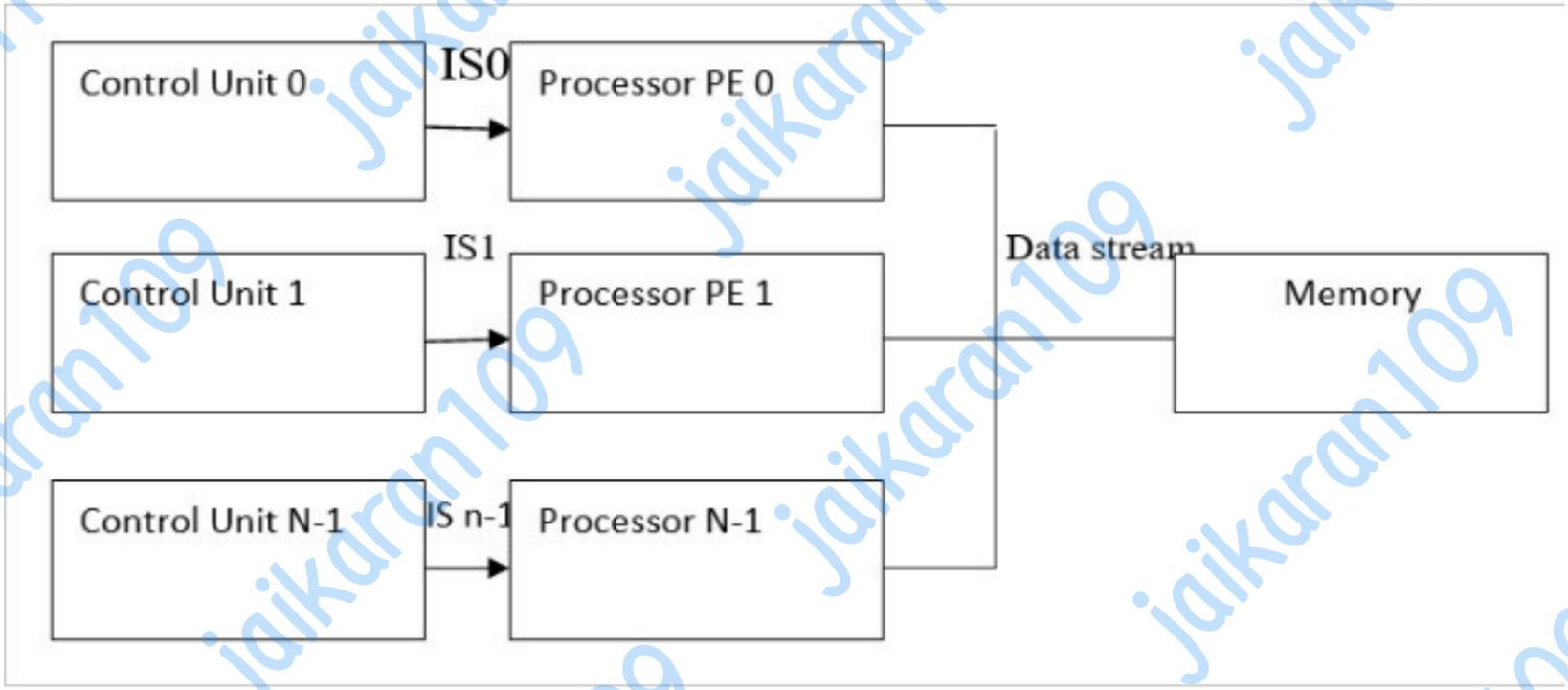


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Flynn's classification:

- **Multiple instruction stream, single data stream (MISD):**

An MISD computing is a multiprocessor machine capable of executing different instructions on processing elements but all of them operating on the same data set.

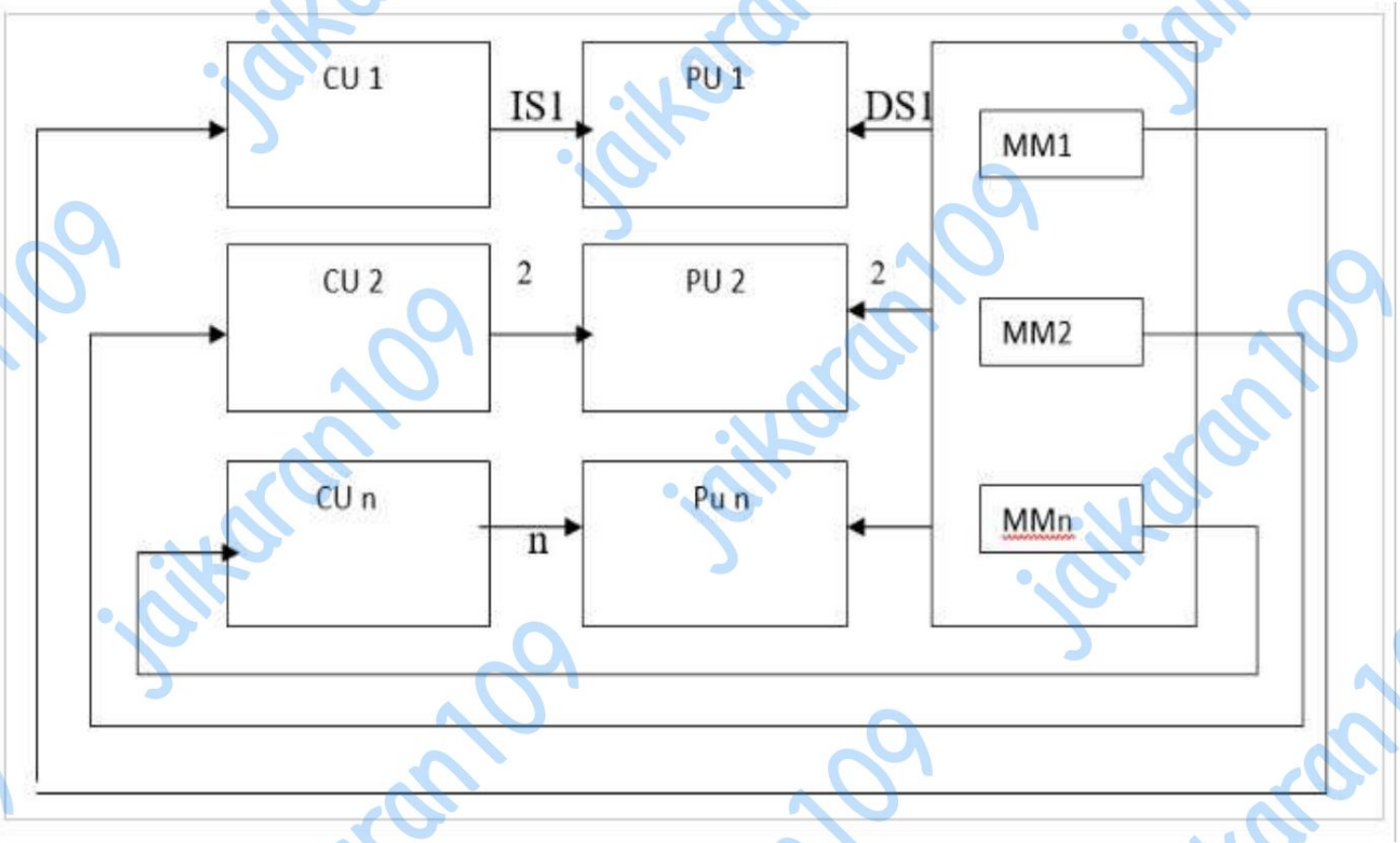


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Flynn's classification:

- **Multiple instruction stream, multiple data stream (MIMD):**

A MIMD system is a multiprocessor machine that is capable of executing multiple instructions over multiple data streams. Each processing element has a separate instruction stream and data stream. **Example:** Distributed computers



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Parallel Processing:

- Flynn's classification depends on the distinction between the performance of the control unit and the data-processing unit. It emphasizes the behavioral characteristics of the computer system rather than its operational and structural interconnections.
- One type of parallel processing that does not fit Flynn's classification is pipelining. The only two categories used from this classification are SIMD array processors and MIMD multiprocessors presented.
- We consider parallel processing under the following main topics:
    1. Pipeline processing
    2. Vector processing
    3. Array processors
- **Pipeline processing** is an implementation technique where arithmetic suboperations or the phases of a computer instruction cycle overlap in execution.
- **Vector processing** deals with computations involving large vectors and matrices.
- **Array processors** perform computations on large arrays of data.

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THANK YOU



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